

Project Summary: Telecom Customer Churn Analysis

I have completed a data analysis project on **Telecom Customer Churn** using a dataset of **7,043 customer records** and **21 Columns**.

The objective of this project was to analyze the customer data and identify **why customers are leaving the service (churn)**. Through data exploration and visualization, I derived several key insights and reasons behind customer churn.

What I Analyzed:

- Percentage of customers who churned
 - Impact of customer **tenure (months of service)**
 - Effect of different **contract types**
 - Influence of **senior citizen status**
 - Role of different **internet and phone services**
 - Comparison of **payment methods**
 - Availability of **security and backup services**
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Key Insights and Findings

1. Churn Rate

Out of 7,043 customers, about 1,869 customers churned — a churn rate of **26.54%**. The remaining 73.46% (approx. 5,174 customers) stayed with the company.

2. Tenure

Customers who stayed for just **1–2 months** had the highest churn rate. Churn decreased as **tenure increased**. Customers with tenure of over 12 months were more loyal.

3. 📄 Contract Type

Month-to-month users churned the most.

Users with **1-year or 2-year contracts** had lower churn rates.

4. 🧓 Senior Citizens

About **16%** of customers were senior citizens.

They showed a **higher churn rate** than non-seniors.

5. 🌐 Services Used

Customers using **Phone Service**, **DSL Internet**, and **Online Security** were more likely to stay.

Churn was higher when **Online Backup**, **Tech Support**, or **Streaming TV** were not used.

6. 💳 Payment Method

Customers using **Electronic Check** had the highest churn rate.

Credit Card and **Bank Transfer** users were less likely to churn.

✅ Final Conclusion

To reduce churn, companies should:

- Promote **long-term contracts**
- Offer **value-added services** (like Online Security & Tech Support)
- Limit the use of **Electronic check** as a payment method
- Focus on **new users** and **senior citizens** for retention

Using these strategies, a company can improve **customer loyalty** and **revenue**.